Table 1 数据库检索

数据库名称（缩写）	网站/访问链接	参考文献（及影响因子 IF）
TOXRIC	https://toxric.bioinforai.tech/home	Wu L, Yan B, Han J, Li R, Xiao J, He S, Bo X. TOXRIC: a comprehensive database of toxicological data and benchmarks. Nucleic Acids Res. 2023;51(D1):D1432-D1445. (Wu 2023 NAR) IF: 13.1
ToxCast	https://www.epa.gov/chemical-research/exploring-toxcast-data-downloadable-data	U.S. EPA. Data download year. Data set name from database version. Retrieved from https://www.epa.gov/chemical-research/toxicity-forecaster-toxcasttm-data on date retrieved. Data release date.(根据使用到的数据在线引用，官网未发现原文 citation)
DILIrank	https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb/drug-induced-liver-injury-rank-dilirank-dataset	Chen M, et al. DILIrank 2.0: An updated and expanded database for drug-induced liver injury risk based on FDA labeling and a literature review. Drug Discov Today. 2025 Nov;30(11):104485. (Chen 2025 Drug Discov Today) IF=7.5
CTD	https://ctdbase.org/	Mattingly CJ, et al. The Comparative Toxicogenomics Database (CTD): a centrally organized resource of chemical-gene/protein interactions, and chemical- and gene-disease relationships. Nucleic Acids Res. 2006 Jan 1;34(Database issue):D490-4. (Mattingly 2006 NAR) IF=4.1
ChEMBL	https://www.ebi.ac.uk/chembl/	Barbara Zdrazil, et al. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. Nucleic Acids Res. 2024 Jan 5;52(D1):D1180 – D1192.(Barbara 2024 Nucleic Acids Res) IF: 13.1

本次纳入说明：

1. 纳入条目及其规则尚未知，以毒性成分、不良反应类别（如肝毒性等）为参考。后期可借鉴多数据库，灵活加入“Target”“Molecular Mass”等。
2. 以 TOXRIC 为例，数据集文件（所有靶器官毒性数据，并选择性纳入 Cardiotoxicity“致癌”和 Mutagenicity“致突变”的数据集，其他尚待定），见下表。
3. 仅初步尝试，尚有许多不足。

Table 2 数据集纳入

原始数据集	毒性类别
Cardiotoxicity_Cardiotoxicity-1.csv	Cardiotoxicity
Cardiotoxicity_Cardiotoxicity-5.csv	Cardiotoxicity
Cardiotoxicity_Cardiotoxicity-10.csv	Cardiotoxicity
Cardiotoxicity_Cardiotoxicity-30.csv	Cardiotoxicity
Endocrine Disruption_NR-AhR.csv	Endocrine Disruption
Endocrine Disruption_NR-AR.csv	Endocrine Disruption
Endocrine Disruption_NR-AR-LBD.csv	Endocrine Disruption
Endocrine Disruption_NR-aromatase.csv	Endocrine Disruption
Endocrine Disruption_NR-ER.csv	Endocrine Disruption
Endocrine Disruption_NR-ER-LBD.csv	Endocrine Disruption
Endocrine Disruption_NR-PPAR-gamma.csv	Endocrine Disruption
Endocrine Disruption_SR-ARE.csv	Endocrine Disruption
Endocrine Disruption_SR-ATAD5.csv	Endocrine Disruption
Endocrine Disruption_SR-HSE.csv	Endocrine Disruption
Endocrine Disruption_SR-MMP.csv	Endocrine Disruption
Endocrine Disruption_SR-p53.csv	Endocrine Disruption
Hepatotoxicity_Hepatotoxicity.csv	Hepatotoxicity
Mutagenicity_Ames Mutagenicity.csv	Mutagenicity
Carcinogenicity_Carcinogenicity.csv	Carcinogenicity
Respiratory Toxicity_Respiratory Toxicity.csv	Respiratory Toxicity

Figure 3 数据清洗结果（展示前十行）：

TAID	Name	IUPAC Name	PubChem CID	Canonical SMILES	InChIKey	Toxicity distribution
TOX-1000	pantoprazole	6-(difluoromethoxy)-2-[(3,4-dimethoxypyridin-2-yl)methylsulfinyl]-1H-benzimidazole	4679	COc1ccnc(CS(=O)c2nc3ccc(OC(F)F)cc3[nH]2)c1OC	IQPSEEYGBUAQFF-UHFFFAOYSA-N	Hepatotoxicity
TOX-10009	Clemizole	1-[(4-chlorophenyl)methyl]-2-(pyrrolidin-1-ylmethyl)benzimidazole	2782	Clc1ccc(Cn2c(CN3CCCC3)nc3ccccc32)cc1	CJXAEXPPLWQRFR-UHFFFAOYSA-N	Cardiotoxicity|Endocrine Disruption
TOX-10016	Thiofentanyl	N-phenyl-N-[1-(2-thiophen-2-ylethyl)piperidin-4-yl]propanamide	62380	CCC(=O)N(c1ccccc1)C1CCN(CCc2cccs2)CC1	YMRFZDHYDKZXPA-UHFFFAOYSA-N	Endocrine Disruption
TOX-10017	NA	11-methoxy-6,8,16,20-tetraoxapentacyclo[10.8.0.02,9.03,7.013,18]icosa-1,4,9,11,13(18)-pentaene-17,19-dione	14421	COc1cc2c(c3oc(=O)c4c(c13)CCOC4=O)C1C=COC1O2	XWIYFDMXXLINPU-UHFFFAOYSA-N	Carcinogenicity|Mutagenicity
TOX-10018	DIMEFLINE	8-[(dimethylamino)methyl]-7-methoxy-3-methyl-2-phenylchromen-4-one	3078	COc1ccc2c(=O)c(C)c(-c3ccccc3)oc2c1CN(C)C	ZXFQRFXLFWWKLX-UHFFFAOYSA-N	Endocrine Disruption
TOX-1002	NA	2-methoxyethanol	8019	COCCO	XNWFRZJHXBZDAG-UHFFFAOYSA-N	Endocrine Disruption|Mutagenicity|Reproductive
TOX-10023	Dodecyl gallate	dodecyl 3,4,5-trihydroxybenzoate	14425	CCCCCCCCCCCCOC(=O)c1cc(O)c(O)c(O)c1	RPWFJAMTCNSJKK-UHFFFAOYSA-N	Endocrine Disruption

TOX-1004	Erlotinib	N-(3-ethynylphenyl)-6,7-bis(2-methoxyethoxy)quinazolin-4-amine	176870	C#Cc1cccc(Nc2ncnc3cc(OCCOC)c(OCCOC)cc23)c1	AAKJLRGGTJKAMG-UHFFFAOYSA-N	Endocrine Disruption|Hepatotoxicity
TOX-1005	rofecoxib	3-(4-methylsulfonylphenyl)-4-phenyl-2H-furan-5-one	5090	CS(=O)(=O)c1ccc(C2=C(c3ccccc3)C(=O)OC2)cc1	RZJQGNCSTQAWON-UHFFFAOYSA-N	Endocrine Disruption|Hepatotoxicity

数据清洗思路：

TAID 在 TOXRIC 数据库是唯一的。首先找出每一个 TAID（如 TOX-1005）在各个数据集的分布，如果一个 TAID 在多个数据集多次出现，说明这个化学物质在多个器官系统中都有毒性，所以多加一列 Toxicity distribution，把毒性分布写出来，用“|”分隔。如“Endocrine Disruption|Hepatotoxicity”。

源代码提供如下：

```
rm(list = ls()) #
setwd("E:\\11_AI drug\\TOXRIC")
# 加载必要的包
library(dplyr)
library(readr)

# 读取所有数据集并添加毒性类型

car <- read_csv("Carcinogenicity_Carcinogenicity.csv")
car$Toxicity_Type <- "Carcinogenicity"
car <- car %>% filter(`Toxicity Value` == 1)

dev_tox <- read_csv("Developmental and Reproductive Toxicity_Developmental Toxicity.csv")
dev_tox$Toxicity_Type <- "Developmental"
dev_tox <- dev_tox %>% filter(`Toxicity Value` == 1)

```

```
rep_tox <- read_csv("Developmental and Reproductive Toxicity_Reproductive Toxicity.csv")
rep_tox$Toxicity_Type <- "Reproductive"
rep_tox <- rep_tox %>% filter(`Toxicity Value` == 1)

Carcinogenicity_Carcinogenicity <- read_csv("Carcinogenicity_Carcinogenicity.csv")
Carcinogenicity_Carcinogenicity$Toxicity_Type <- "Carcinogenicity"

Cardiotoxicity_Cardiotoxicity_1 <- read_csv("Cardiotoxicity_Cardiotoxicity-1.csv")
Cardiotoxicity_Cardiotoxicity_1$Toxicity_Type <- "Cardiotoxicity"

Cardiotoxicity_Cardiotoxicity_5 <- read_csv("Cardiotoxicity_Cardiotoxicity-5.csv")
Cardiotoxicity_Cardiotoxicity_5$Toxicity_Type <- "Cardiotoxicity"

Cardiotoxicity_Cardiotoxicity_10 <- read_csv("Cardiotoxicity_Cardiotoxicity-10.csv")
Cardiotoxicity_Cardiotoxicity_10$Toxicity_Type <- "Cardiotoxicity"

Cardiotoxicity_Cardiotoxicity_30 <- read_csv("Cardiotoxicity_Cardiotoxicity-30.csv")
Cardiotoxicity_Cardiotoxicity_30$Toxicity_Type <- "Cardiotoxicity"

Endocrine_Disruption_NR_AhR <- read_csv("Endocrine Disruption_NR-AhR.csv")
Endocrine_Disruption_NR_AhR$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_NR_AR <- read_csv("Endocrine Disruption_NR-AR.csv")
Endocrine_Disruption_NR_AR$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_NR_AR_LBD <- read_csv("Endocrine Disruption_NR-AR-LBD.csv")
Endocrine_Disruption_NR_AR_LBD$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_NR_AR_aromatase <- read_csv("Endocrine Disruption_NR-aromatase.csv")
Endocrine_Disruption_NR_AR_aromatase$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_NR_ER <- read_csv("Endocrine Disruption_NR-ER.csv")
Endocrine_Disruption_NR_ER$Toxicity_Type <- "Endocrine Disruption"
```

```
Endocrine_Disruption_NR_ER_LBD <- read_csv("Endocrine Disruption_NR-ER-LBD.csv")
Endocrine_Disruption_NR_ER_LBD$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_NR_PPAR_gamma <- read_csv("Endocrine Disruption_NR-PPAR-gamma.csv")
Endocrine_Disruption_NR_PPAR_gamma$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_SR_ARE <- read_csv("Endocrine Disruption_SR-ARE.csv")
Endocrine_Disruption_SR_ARE$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_SR_ATAD5 <- read_csv("Endocrine Disruption_SR-ATAD5.csv")
Endocrine_Disruption_SR_ATAD5$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_SR_HSE <- read_csv("Endocrine Disruption_SR-HSE.csv")
Endocrine_Disruption_SR_HSE$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_SR_MMP <- read_csv("Endocrine Disruption_SR-MMP.csv")
Endocrine_Disruption_SR_MMP$Toxicity_Type <- "Endocrine Disruption"

Endocrine_Disruption_SR_p53 <- read_csv("Endocrine Disruption_SR-p53.csv")
Endocrine_Disruption_SR_p53$Toxicity_Type <- "Endocrine Disruption"

Hepatotoxicity_Hepatotoxicity <- read_csv("Hepatotoxicity_Hepatotoxicity.csv")
Hepatotoxicity_Hepatotoxicity$Toxicity_Type <- "Hepatotoxicity"

Mutagenicity_Ames_Mutagenicity <- read_csv("Mutagenicity_Ames Mutagenicity.csv")
Mutagenicity_Ames_Mutagenicity$Toxicity_Type <- "Mutagenicity"

Respiratory_Toxicity_Respiratory_Toxicity <- read_csv("Respiratory Toxicity_Respiratory Toxicity.csv")
Respiratory_Toxicity_Respiratory_Toxicity$Toxicity_Type <- "Respiratory Toxicity"

# 合并所有数据集
combined_tox <- bind_rows(
```

```
  dev_tox, rep_tox, Carcinogenicity_Carcinogenicity,
  Cardiotoxicity_Cardiotoxicity_1, Cardiotoxicity_Cardiotoxicity_5,
  Cardiotoxicity_Cardiotoxicity_10, Cardiotoxicity_Cardiotoxicity_30,
  Endocrine_Disruption_NR_AhR, Endocrine_Disruption_NR_AR,
  Endocrine_Disruption_NR_AR_LBD, Endocrine_Disruption_NR_AR_aromatase,
  Endocrine_Disruption_NR_ER, Endocrine_Disruption_NR_ER_LBD,
  Endocrine_Disruption_NR_PPAR_gamma, Endocrine_Disruption_SR_ARE,
  Endocrine_Disruption_SR_ATAD5, Endocrine_Disruption_SR_HSE,
  Endocrine_Disruption_SR_MMP, Endocrine_Disruption_SR_p53,
  Hepatotoxicity_Hepatotoxicity, Mutagenicity_Ames_Mutagenicity,
  Respiratory_Toxicity_Respiratory_Toxicity
)

# 移除 Toxicity Value 列（如果存在）
if("Toxicity Value" %in% colnames(combined_tox)) {
  combined_tox <- combined_tox %>% select(-`Toxicity Value`)
}

# 处理重复的 TAID，创建毒性分布列
tox_summary <- combined_tox %>%
  group_by(TAID) %>%
  summarise(
    Name = first(na.omit(Name)),
    `IUPAC Name` = first(na.omit(`IUPAC Name`)),
    `PubChem CID` = first(na.omit(`PubChem CID`)),
    `Canonical SMILES` = first(na.omit(`Canonical SMILES`)),
    InChIKey = first(na.omit(InChIKey)),
    `Toxicity distribution` = paste(sort(unique(Toxicity_Type)), collapse = "|")
  ) %>%
  ungroup()

# 保存整理后的数据到新的 CSV 文件
write_csv(tox_summary, "Combined_Toxicity_Data.csv")
```

```

# 显示处理结果摘要
cat("数据处理完成! \n")
cat("合并后唯一 TAID 数量:", nrow(tox_summary), "\n")
cat("具有多重毒性的化合物数量:", sum(grepl("\\|", tox_summary$`Toxicity distribution`)), "\n")
cat("毒性类型分布:\n")
table(unlist(strsplit(tox_summary$`Toxicity distribution`, "\\|")))

```